21.14. BENEFICIAL UTILIZATION OF NUCLEAR WASTES

The amount of spent fuel generated during 2003 amounts to about 8000 t. It contains large and potentially valuable sources of metals and radioactive nuclides. The known and predicted cumulative amounts are indicated in Figure 21.11. Though today considered a liability it may in the future become a needed asset. Since the extraction and utilization of some of the fission products or actinides will probably not be economic after the waste has been vitrified and placed in permanent geologic storage, the nuclear fuel reprocessing scheme should therefore be designed for byproduct extraction. Presently the most interesting products in the waste are the platinum group metals (due to their metal values) and ^{90}Sr, ^{137}Cs, ^{85}Kr, ^{237}Np/^{238}Pu, and ^{241}Am (due to their radiation properties, cf. §§8.11, 18.5, etc).

The waste contains considerable amounts of Ru, Rh, and Pd, all metals in scarce abundance on earth. These elements are used as catalysts in the chemical industry, for catalytic exhaust cleaning in cars, and as corrosion resistant materials. The United States demand exceeds the domestic production by about a factor of 100. The United States would be independent of import from year 2000 if these elements were recovered from the yearly generated spent fuel. This is particularly true if technetium is recovered, since it can often replace platinum. Some of the recovered elements would be radioactive, but the activities would be small enough to make the elements easy to handle.

Beta radiation from ^{85}Kr on phosphors causes visible light. Radiokrypton light sources have widespread applications where reliable lights are required as, for example, at airports, railroads, hospitals, etc., or where sources of electricity could cause dangerous explosions, as in coal mines, natural gas plants, etc. However, fission product krypton contains only 4% of ^{85}Kr, which makes it unsuitable for high intensity lightning applications. ^{85}Kr must therefore be enriched about a factor of 10, which presently can be done e.g. by thermal diffusion.

By the year 2000, over 100 MW heat will be produced by radiostrontium. Strontium fuelled thermoelectric generators are used in several countries to power unmanned weather data acquisition systems, lighthouses, and other navigation aids, etc. Their reliability surpasses any other remote power source. The current thermoelectric generators have a thermal-to-electrical efficiency of about 5%, while recently developed thermal-to-mechanical systems show efficiencies of 25–30%. The use of such systems could be expanded vastly by increased recovery of suitable fission products and actinides. However, due to environmental concerns many of the generators deployed so far at sea or on shores are now being recovered and scrapped.

Food sterilization by radiation is potentially of global importance (§8.11). Though this presently is done by ^{60}Co or accelerator radiation, the advent of large quantities of radiocesium recovered from nuclear wastes may have a considerable positive impact on the economics and scale of food irradiation. An almost equally important use of ^{137}Cs

would be for sewage sludge treatment, which may become increasingly important as the requirements for sterilization and secondary treatment of the sludge increases. The thermoradiation of sludge may also make it useful as a sterile fertilizer.

The low penetrating radiation, long half-life, and high power density of ^{238}Pu makes it ideal for special purpose power supplies (§7.9.3). The main present use is in space research, and $\sim$30 kW_{th} power sources have been launched into space, ^{238}Pu has been used in heart pacemakers and is still a candidate as a power source for completely artificial hearts. Production of ^{238}Pu requires the isolation of ^{237}Np, which is then irradiated and reprocessed to produce pure ^{238}Pu.

The demand for ^{241}Am is larger than present production capacity because of its use in logging oil wells, in smoke detectors, and for various gauging and metering devices. However, the potential source is large. At the end of 2003, the amount of americium produced as a result of civil nuclear power reactor operation was about 87 tonnes. About 68 tonnes of this americium resulted from the decay of ^{241}Pu subsequent to its discharge from the reactors. Currently, the americium inventory increases by about 4 tonnes per year.

If all the mentioned waste products are recovered, this would mean (a) that the waste is turned into an essential asset with benefits in food production, health, and safety, and (b) that the hazard of the remaining waste would be much lower, considerably simplifying final waste storage.

21.15. EXERCISES

21.1. In a BWR the minimum and maximum heat fluxes at the fuel rod surface are 0.46 and 1.08 MW m^{-2} at a coolant temperature of 283°C. The rods have an outer diameter of 12.7 mm with a cladding of 0.8 mm thick Zircaloy-4; assume a negligible fuel cladding gap and neglect the temperature drop between coolant and cladding and across the fuel-cladding gap. Assume that data in Table 21.1 are valid at all temperatures and $k_{UO2} = 4$ W/m°C. What are the highest and lowest fuel-center temperatures?

21.2. Using the data for the Würgassen reactor and the thermal neutron capture cross-section (Table 19.4) it can be calculated how many kg Pu should be formed per t U at a burn-up of 27 500 MWd/t. (a) Make this calculation assuming that plutonium disappears only through fission in ^{239}Pu. (b) According to Table 21.2 each t U from a PWR contains 8.69 kg Pu; why is your result much lower?

21.3. It is desired that 98% of all ^{233}Th formed by neutron capture in ^{232}Th decays to ^{233}U. How long a time must elapse between end of irradiation and start of reprocessing?

21.4. In a ^{233}U fuelled reactor, some ^{233}U is converted into ^{235}U. Calculate the amount of ^{235}U formed in 1 t ^{233}U from neutron capture in ^{233}U and ^{234}U ($\sigma_{n,\gamma}$